

User Manual

IEC 61850 SV, GOOSE, PTP, SCL, and PCAP evidence workflow

Version	1.4.0-beta.2
Community license	GPL-3.0-or-later
Platform	Windows 10/11 x64
Product boundary	Receive-only; no SV/GOOSE publishing or IEC 61850 control commands

This document describes engineering use and evidence boundaries. It is not a certificate of IEC 61850 conformance, calibrated timing, functional safety, cybersecurity approval, equipment isolation, or switching authority.

Product boundary

- Process Bus Insight is receive-only and raw-passive.
- It does not publish Sampled Values or GOOSE and does not send IEC 61850 control commands.
- It complements—not replaces—approved test equipment, project procedures, and engineering authority.

Sampled Values workspace

- Use stream identity, APPID, svID, VLAN, source MAC, confRev, counters, waveform, RMS, phasor, and sequence diagnostics for the selected stream.
- The selected stream owns the displayed waveform, metering, phasor, and details. Cross-stream combinations are not valid evidence.
- Missing, duplicate, gap, or timing indications are investigation leads whose confidence depends on the capture path.

GOOSE workspace

- Inspect publisher identity, stNum, sqNum, typed dataset values, and event history.
- Observed frames do not prove that an external IED accepted, trusted, or acted on the traffic.

PTP timing context

- Review visible transport, domain, message type, grandmaster context, and freshness where decoded.
- The application does not discipline clocks or certify time synchronization.

SCL expected-vs-observed

- Load SCD, ICD, or CID files and compare expected APPID, destination MAC, VLAN, DataSet, svID, confRev, and related identity evidence.
- A match describes evidence at the selected observation point; it is not formal conformance certification.

PCAP replay

- Classic Ethernet PCAP can be replayed through the same decoder/analyzer entry point used by live capture.
- Replay preserves recorded evidence but cannot recreate switch loading, NIC buffering, unrecorded packet loss, or hardware timestamp behavior.

Security and data handling

- Treat captures and engineering files as potentially sensitive.
- Sanitize customer, employer, station, device, credential, MAC/IP, and project identifiers before sharing.
- Report parser or package-integrity vulnerabilities through a private GitHub security advisory.

Licensing and source

- Post-transition community revisions and packages use GPL-3.0-or-later.
- The historical Apache-2.0 boundary is commit 85d43a0fe58a5888a9e8008c168ab76d2333ea87.

- SOURCE.md identifies the exact source corresponding to a packaged binary. The commercial notice is not itself a license.

Evidence wording

Use terms such as configured, observed, decoded, calculated, replayed, screened, provisional, unsupported, or not yet verified. Do not claim certification, calibration, deterministic timing, functional safety, cybersecurity approval, universal interoperability, switching authority, or IED acceptance without separate competent evidence.